Havalyn Hale

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EXPERIENCE

Olin Electric Motorsports September 2026 - Present
Low Voltage Engineer *Needham, MA*

- Revamping and finishing the Formula SAE car's telemetry system: Python on a Raspberry Pi (Linux) relaying CAN bus data to a ground station over 915 MHz SiK Holybro radios, with cross-vehicle support planned for MkVII and MkVIII
- Reconfigured the radios from 9600 to 57600 baud and redesigned the message format, cutting each packet from ~1,870 bytes to ~40 and raising throughput from ~3 to 100+ messages/sec; added per-ID rate limiting
- Designing the ground station for a ~700 m track; a worst-case link model at 1 km predicts a 29 dB margin

Olin Vine Robotics August 2026 - Present
Research Assistant *Needham, MA*

- Independent study on the mechanics and modeling of vine-inspired soft growing robots (advisor: John Bell)
- Co-designed a steering control board with 6 independent pressure channels, each with a regulator, gauge, pressure sensor, and vent valve, giving the robot 6 DOF; building an expandable air manifold with 2 DOF of steering now and a path to 5+
- Wired a Raspberry Pi Pico sensing board that reads a Honeywell ABP sensor's raw 14-bit output over I²C, masking out the status bits; wrote embedded C firmware that rejects bad readings and zeroes the sensor at startup, bringing noise down to a few thousandths of a psi
- Building a second sensing board, running alongside the steering board, that reads 6 Honeywell ABP 0-5 psi sensors through a multiplexer across three robot sections

Olin Plasma Engineering Lab (OPEL) January 2026 - Present
Researcher *Needham, MA*

- Co-developed MATLAB models for the MK5 Hall Effect Thruster on an 8 kg CubeSat: sized propellant (0.11-0.22 kg for a 280 m/s apogee raise across 10-20 km/s exhaust velocities) and predicted approximately 15 mN thrust, 1,630 s Isp, and 305 W at 0.95 mg/s argon flow; compiled results as an Olin ISR
- Co-developed design requirements for a 3U CubeSat-scale thruster; performed thin-wall stress analysis of an Al 6061-T6 propellant tank (3 mm wall, 14.3 MPa), yielding a 1.08 safety factor

Olin Shop August 2026 - Present
Electronics Assistant *Needham, MA*

- Redesigned soldering training to add de-soldering techniques and streamline delivery; sessions cover component identification, joint technique, safety, and schematic/bill of materials reading

Olin College of Engineering September 2026 - Present
Academic Resource Co-Advisor (ARC) *Needham, MA*

- Provide 1:1 and walk-in academic support on time management, study strategies, and stress management, using training in facilitation, academic coaching, and peer mentorship

PROJECTS

Ground Station RF Link Model (Olin Electric Motorsports)

Researched free-space path loss and the two-ray ground reflection model, worked the link budget by hand, then implemented it in Python with NumPy; modeled a 1 km, 915 MHz link predicting a 29.6 dB margin against a 20 dB target

Python, NumPy

Rocky Animatronic (PIE, team project)

Building a 4-legged walking robot (12 VS-2HMB servos, 3 per leg) on an Arduino Nano as the MVP for an animatronic of Rocky from Project Hail Mary, with a fifth leg planned as an arm

Wired and labeled the servo driver; writing C++ PWM code with the Adafruit library to implement the walking gait; diagnosing current draw and servo stall issues

C++, Arduino Nano

EDUCATION

B.S. in Electrical and Computer Engineering *Olin College of Engineering* Expected May 2029
Needham, MA

4-Year 50% Merit Tuition Scholarship. GPA: 4.0. Relevant Coursework: Software Design, Discrete Mathematics (in progress), Modeling and Simulation, Quantitative Engineering Analysis, Instrumentation Sensors and Measurement

SKILLS

Programming: Python (NumPy), C, C++, MATLAB, Java

Embedded & Systems: Linux/Ubuntu, Embedded Systems, Raspberry Pi, CAN Bus, Serial Communication (pyserial), I²C, PWM, Arduino, RF Link Budgets

Tools & Lab: GitHub, Altium Designer, KiCAD, SOLIDWORKS, COMSOL, Oscilloscope, Logic Analyzer, Soldering/De-soldering, 3D Printing